

JENNY LEOPOLDINA SMITH, MSc

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Bioinformatician with over 9 years experience analyzing NGS data, evaluating sequencing performance and QC metrics, as well as validating analytical methods. Develops analysis pipelines for accuracy, reproducibility, and robustness. Commitment to generating documentation for analytical, statistical, and benchmarking for technical reports. Collaborates across diverse interdisciplinary teams to advance cancer genomics research.

Core Skills

- **NGS analysis & sequencing QC:** evaluation of sequencing performance and quality-control metrics; outlier and anomaly detection across samples and batches; primary, secondary and tertiary analyses of WGS, RNA-seq, and single-cell data (FASTQ, BAM, VCF, GTF)
- **Algorithm & pipeline validation:** benchmarking and selection of computational methods; end-to-end pipeline validation for accuracy and reproducibility; nf-test, unit testing, CI/CD
- **Programming:** Python (pandas, scripting and CLI tools, venv), R (Tidyverse, Bioconductor, package development, renv), Bash/Linux
- **Variant analysis:** SNV, CNV, and structural-variant calling (somatic and germline); annotation and clinical filtering (bcftools, AnnotSV, Ensembl VEP; gnomAD, ClinVar, COSMIC)
- **Pipelines, containers & cloud:** Nextflow DSL2, nf-core, Snakemake; Docker, Singularity, Apptainer; AWS (Batch, S3, EC2), Azure, Google Cloud HPC; SLURM / PBS / HyperQueue
- **Documentation & collaboration:** technical reports, SOPs, and contributing guidelines; Git/GitHub, Confluence, Jira; cross-functional work with wet-lab, clinical, and quality teams; SQL (SQLite, PostgreSQL — strengthening)

Experience

Computational Biologist, Consultant, *Ospedale Pediatrico Bambino Gesù, Rome, Italy* June 2025 – Present

- Benchmarked and validated structural-variant calling algorithms—Severus, SAVANA, and the MINDA ensemble—for long-read WGS, then integrated the selected approach into a somatic-variant annotation and clinical-filtering pipeline for a retrospective cohort of more than 60 primary patient DNA samples.
- Built automated QC and reporting workflows using Quarto, IGV, and JSON to evaluate sequencing outputs, identify outlier samples and batch-level inconsistencies, and support interactive data exploration through a Quarto dashboard.
- Developed reproducible Nextflow DSL2 and Snakemake pipelines (>100 TB NGS data), employing version control (git and github), containerization, and automated CI/CD testing to ensure analyses remain accurate and robust as they scale.
- Authored technical reports, study summaries, and SOPs; initiated bi-weekly cross-functional meetings with molecular biologist and clinical collaborators to align data processing with analysis milestones.

Computational Biologist, Senior, *Seattle Children's Research Institute, Seattle, WA* April 2022 – September 2024

- Developed and adapted reproducible Nextflow and nf-core pipelines for RNA-seq, Cut&Run, and ATAC-seq with and CI/CD testing; released documented public workflows on the SCRI GitHub organization.
- Supported quality and troubleshooting for research teams by leading office hours, and debugging custom and off-the-shelf software. Built parameterized reporting templates, to facilitate generation of statements of work (SOWs) and internal documentation for new HPC users.
- Analyzed multi-sample scRNA-seq and scATAC-seq (Scanpy, Seurat, CellRanger) with QC, dimension reduction, and cell-type classification.

Bioinformatics Analyst II, *Fred Hutchinson Cancer Center, Seattle, WA* January 2017 – April 2022

- Developed custom Nextflow RNA-seq pipelines on AWS (Batch, S3, EC2) and on-premises HPC with automated QC to process more than 2,500 samples reproducibly; data outputs contributed to 10+ peer-reviewed publications.
- Developed expression-based classifiers and prognostic signatures for pediatric leukemia risk stratification using supervised and unsupervised machine learning.
- Built an internal R package (Roxygen/usethis) and a team GitHub organization for reproducible biomarker discovery from clinical and expression data stored in SQL databases.

Postbaccalaureate Fellow, *National Institutes of Health (NIAMS), Bethesda, MD* June 2014 – June 2016

- Applied bioinformatics methods to analyze ATAC-seq and ChIP-seq datasets, identifying gene regulatory mechanisms underlying human keratinocyte development.
- Completed six advanced bioinformatics courses through the Foundation for Advanced Education in the Sciences (FAES) and National Library of Medicine (NLM).

Selected Software & Pipeline Projects

- **WGS Nanopore SV annotation pipeline:** Nextflow DSL2 workflow that merges, annotates, and filters somatic structure and single-nucleotide variants against public databases (gnomAD, ClinVar, VEP) and supports custom cancer domain-specific reference (BED) files for annotation; samples' results are aggregated across the cohort into a single parameterized, interactive Quarto HTML report. Providing robust and accurate structural variant interpretation and prioritization for retrospective clinical studies.
- **[RNA-seq quantification & QC pipeline \(rnaseq_count_nf\)](#):** Scalable Nextflow DSL2 workflow that reproducibly processes samples in parallel (STAR, RSeQC, Picard, MultiQC), containerized software, and automated CI/CD testing.
- **[Pipeline contributing guidelines](#):** Authored team standards and reporting templates covering version control, code review, data management, and documentation for reproducible, auditable analytical deliverables.

Education

- **Applied Bioinformatics and Genomics, M.Sc.**, University of Oregon, Eugene, OR
- **Secondary Education in Science, M.Ed.**, Arizona State University, Phoenix, AZ
- **Biology, B.A.**, *magna cum laude*, University of San Diego, San Diego, CA

Selected Publications

- Zanghi G.*, Patel H.*, **Smith J.L.***, ... Kappe S.H.I. Genome-wide gene expression profiles throughout human malaria parasite liver-stage development in humanized mice. *Nature Microbiology*. 2025.
- Farrar J.E., **Smith J.L.**, Othus M., ... Meshinchi S. Long noncoding RNA expression independently predicts outcome in pediatric acute myeloid leukemia. *Journal of Clinical Oncology*. 2023.
- Huang B.J., **Smith J.L.**, Farrar J.E., ... Meshinchi S. Integrated stem cell signature and cytomolecular risk determination in pediatric acute myeloid leukemia. *Nature Communications*. 2022.
- **Smith J.L.**, Ries R.E., Hylkema T., ... Meshinchi S. Comprehensive transcriptome profiling of cryptic CBFA2T3-GLIS2 fusion-positive AML. *Clinical Cancer Research*. 2019.

* co-first author.

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Continuing Education & Community

- 2025 Relational Databases and SQL, freeCodeCamp / project-based
- 2024 DevOps for Data Scientists, POSIT Conference, Seattle, WA
- 2024 Nextflow nf-core Hackathon, University of Washington, Seattle, WA
- 2022 – 2024 Data Science Learning Community member, DSLC.io
- 2018–2023 R-Ladies event organizer, Seattle, WA